Additionally, there is a deficiency in evaluation approaches that incorporate measures for external supervision during the AI Scientist's implementation process. The deeper issue is that the quality of scientific discovery often lacks unified objective standards, and the process of scientific exploration is filled with uncertainty and openness, making comprehensive evaluation and effective supervision of AI Scientist's verification capability exceptionally difficult. Evaluating AI Scientist's output (e.g., generated papers) from a peer review perspective, while being a results-oriented assessment method, also has inherent limitations. As in human research systems, even experienced peer reviewers may not always accurately identify the groundbreaking and far-reaching work. A frequently cited example is that the word2vec paper (Mikolov et al., 2013) was initially rejected by ICLR 2013, but later received the "Test of Time Award" at NeurIPS 2023. Extensive experimental analyses have demonstrated that review scores are not reliable indicators for predicting future impact (Abramo et al., 2019, Cortes and Lawrence, 2021), suggesting that peer review may be more suitable for filtering low-quality papers rather than identifying the highest quality papers.

5. Ethical Considerations

Sub-Position: *AI scientists are in urgent need of a comprehensive system for generation management and quality evaluation.*

As autonomous research agents, AI Scientists lack values and moral constraints. They are incapable of making ethical judgments about the societal impact of their work, and they do not self-regulate based on potential risks associated with their findings (Bengio et al., 2025). As AI Scientists possess stronger capabilities in idea generation and experiment execution, their influence on scientific research and society could far surpass that of current LLMs and scientific tools (e.g., Deep Search, AutoSurvey (Wang et al., 2024c)). In the absence of proper oversight, **AI Scientists may: (1) be misused**, overwhelming the peer review system, leading to a decline in overall research quality; **(2) enter unethical or dangerous research domains**, autonomously generating and publishing sensitive findings that accelerate the development of harmful technologies; **(3) weaken the quality of PhD training**, leading to a decline in human research standards and overall scientific literacy. To prevent the above situations, **we argue that AI Scientists are in urgent need of a comprehensive system for generation management and quality evaluation**, thus enabling effective behavior regulation within the human moral framework (Jobin et al., 2019). This system should include, but not be limited to, the following components:

(1) Implement measures to prevent AI-generated content from disrupting human review systems: Effective strategies should be adopted to ensure that AI-generated articles do not interfere with human peer-review systems while maintaining high standards of quality. This includes establishing a centralized platform to archive scientific outputs generated by AI Scientists, developing automated detection systems to identify such content, and creating specialized evaluation tools (e.g., DeepReview (Zhu et al., 2025)) to assess the quality of AI-generated research outputs. These tools should help identify and filter low-quality content, thereby reducing the burden on the peer review process. All AI-generated outputs must be transparently labeled and reviewed, including information on their origin, generation methods, and scientific tools.

(2) Establish boundaries and strengthen training programs: Implement clear boundaries between human-led and AI-led research processes to ensure that PhD students receive comprehensive training. Key

components of doctoral education(e.g., idea testing), should prioritize human involvement to maintain high standards of scientific literacy. Additionally, guidelines should be established to prevent over-reliance on AI Scientists in PhD training, ensuring that AI tools serve as supplements rather than substitutes in the educational process.

(3) Formulate an ethics and responsibility convention: A global convention should be established to define the ethical boundaries and risk management principles for AI-driven research (Huang et al., 2022). All researchers and institutions utilizing AI Scientists must fully disclose the generation process, algorithmic sources, training data, and potential societal risks of their findings. Additionally, a hybrid mechanism combining automated and human-in-the-loop review should be implemented for continuous ethical oversight and risk evaluation, ensuring that AI Scientist research activities remain within socially safe boundaries (Jobin et al., 2019, Khan et al., 2022). Furthermore, appropriate legislation should be developed to regulate AI Scientists by imposing strict limitations on their use for specific research purposes.

6. Future Directions

This section outlines feasible pathways to bridge the current implementation capability gap of AI Scientists. Addressing foundational **Basic Abilities** is paramount. While scaling laws for pre-training and post-training (Kaplan et al., 2020, Zhang et al., 2025) promise progressive LLM improvements, immediate strategies like well-defined Workflows (Li et al., 2024d, Gu et al., 2024b) also can mitigate current implementation weaknesses. Structuring research processes with human-defined tools allows for guided AI execution and targeted interventions. For instance, Retrieval Augmented Generation (RAG) can counteract limitations in handling long texts or accessing current information (Fan et al., 2024, Arslan et al., 2024), thus expanding the knowledge scope of AI systems.

A significant challenge for sophisticated **Strategic Planning** is the immense resource consumption of RL (Cao et al., 2024). A promising direction to alleviate this involves leveraging LLMs to simulate aspects of the environment or task execution, thereby accelerating the RL feedback loop (Sun et al., 2025). By allowing the RL agent to receive quicker, albeit potentially approximate, feedback on its actions, particularly for operations that are inherently time-consuming in the real world, the sampling efficiency may be significantly improved. This could reduce the extensive wall-clock time typically required for training robust long-horizon planning and adaptive meta-thinking capabilities in complex scientific domains.

Ensuring **Reliable Verification and Fostering Collaboration** is crucial. Standardized protocols like MCP and A2A (Yang et al., 2025b, Ray, 2025, Hou et al., 2025) can establish basic interoperability. A promising direction is to build modular multi-agent systems, where specialized AI agents for sub-tasks (e.g., literature review, code generation) are coordinated by a central "Planner Agent" trained via advanced RL, leveraging existing tools (e.g., PASA (He et al., 2025)) rather than reinventing capabilities. Furthermore, enhanced oversight of AI Scientist inference processes is imperative, not just to prevent benchmark "hacking", but also to instill ethical boundaries against unscrupulous data acquisition or other problematic behaviors.

Finally, the **Evaluation** of AI Scientists (Chang et al., 2024) must evolve towards a holistic, coarse-grained paradigm reflecting real-world scientific discovery's multifaceted nature. Scientific breakthroughs involve both practical utility and novelty. Thus, evaluation frameworks should go beyond single-metric optimization, adopting multi-objective criteria that assess performance gain, originality, experimental rigor, and communication clarity. This multi-faceted approach will offer a more accurate measure of an AI Scientist's true

contribution, guiding development toward impactful scientific exploration.

7. Conclusion

The rise of AI Scientists marks a paradigm shift in scientific discovery, with large language models (LLMs) now driving the workflow from idea generation to experiment execution. Recent systems have shown promise, producing research accepted at ICLR 2025 workshops and sparking discussions on the imminence of human-level AI Scientists. However, despite this progress, AI Scientists have yet to achieve breakthroughs in computer science comparable to traditional automated tools. Based on benchmark analyses and a systematic evaluation of 28 papers from five leading AI Scientist systems, we identify a core bottleneck: the inability to reliably execute and verify experiments. This implementation gap limits both scientific rigor and the quality of the research output. We analyze its root causes and call on the community to address this critical limitation.

Alternative Views. An alternative viewpoint suggests that AI Scientists need not pursue completely autonomous implementation capabilities in the short term, but rather facilitate human-machine collaboration as Co-scientists to assist humans. This approach avoids the deficiencies of LLMs in Dynamic Planning capabilities and Reliable Verification capabilities, instead allowing AI to focus on its strengths, such as idea generation, while humans execute the specific experimental results (Weng et al., 2025). If an AI system, though unable to independently complete all implementation details, can increase human scientists' efficiency tenfold, or help human scientists conceive and verify complex ideas previously beyond reach, then it undoubtedly also qualifies as a successful "collaborative scientist."

Acknowledgements

The genesis of this position paper traces back to the insightful discussions and interactions at the AI Co-scientist Discussion held in conjunction with ICLR 2025 on April 26, 2024 [1]. We extend our sincere gratitude to the invited speakers, including Chenglei Si, Jindong Wang, Yutaro Yamada, and David Ha, whose perspectives are invaluable. We also deeply appreciate the contributions of the more than 200 participants who engaged in the vibrant discussions on that day; many of the ideas explored in this work were sparked and refined through those collective interactions. We thank every participant for their engagement and for fostering a stimulating environment that significantly shaped our thinking.

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